

Laboratory 1 - Cluster Analysis

Clustering is a set of techniques used to partition data into groups (called clusters). Clusters are loosely defined as groups of data objects that are more similar to other objects in their cluster than they are to data objects in other clusters. Clustering helps identify two qualities of data: meaningfulness and usefulness.

TASKS TO BE CARRIED OUT

1. Cluster using **Agglomerative Clustering** following datasets: 1_S-sets and 4_Nested-datasets; select one file from each directory. Find the correct number of clusters using dendrograms. During clustering, to calculate the distance between two clusters, test the four criteria: single-linkage, complete-linkage, average-linkage and ward. Select the best results for each dataset. Plot the final clusters with different colors. Can we use Agglomerative Clustering for 2_Birch-sets? Discuss the results. - 4 pkt

2. Compare used distance criteria regard to the dataset. - 1 pkt

USEFUL INFORMATION

For **dendrogram** use `scipy.cluster.hierarchy` package.

```
"package".dendrogram("package".linkage(dataset, method = 'ward'))  
# method - criteria to calculate the distance between two clusters  
# method{'ward', 'complete', 'average', 'single'}, default='ward'
```

For **AgglomerativeClustering** use `sklearn.cluster` package.

```
ac = "package".AgglomerativeClustering(n_clusters = 2, affinity = 'euclidean', linkage = 'ward')  
# linkage - criteria to calculate the distance between two clusters  
# linkage{'ward', 'complete', 'average', 'single'}, default='ward'  
# n_clusters - number of clusters
```

```
fit_ac = ac.fit_predict(dataset)  
# Plot obtained clusters  
plt.scatter(dataset[:,0],dataset[:,1],s=10,c=fit_ac[:,],cmap='hsv')  
# where plt.scatter(x-data, y-data, s=size of points, cmap=type of points colors)  
plt.show()
```